

iVFAS: An Improved Vehicle-to-Fog Authentication System for Secure and Efficient Fog-Based Road Condition Monitoring

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Abstract—Road condition monitoring schemes allow a severe reduction of traffic queues and drastically improve road safety. In these schemes, authorized vehicles communicate the current road condition via a Road Side Unit (RSU) or fog node. The reaction time for these schemes greatly improves in case the RSUs are capable of locally determining the validity of the messages and if relatively low-cost cryptographic operations are involved to ensure the required security features like confidentiality, mutual authentication, anonymity, untraceability, etc. The current literature on vehicular ad hoc networks (VANETs) does not cover all these mentioned issues profoundly. We show that a recently outperforming scheme called the V2F authentication scheme (VFAS) suffers from several severe issues like no protection against impersonation, traceability, perfect forward secrecy, and ephemeral leakage attacks, no possibility to revoke certificates, scalability problems, and being vulnerable for a semi-trusted third party. We propose an improved scheme, called iVFAS, to address all the required security features and which has better computational, communication, storage and energy consumption performance with respect to most of the related literature, including VFAS. In addition to this, we also evaluated performance under unknown attacks and practical implementation using NS3. Moreover, this new scheme also includes a variant of the Elliptic Curve Qu Vanstone (ECQV) certificate scheme in the registration phase, which is resistant to the extended Canetti–Krawczyk adversary model.

Index Terms—Vehicle to infrastructure (V2I), road condition monitoring, privacy, mutual authentication, road Side Unit (RSU), fog computing, Elliptic Curve Qu Vanstone (ECQV) certificate scheme, extended Canetti–Krawczyk (eCK) adversary model.

I. INTRODUCTION

INTELLIGENT Transport Systems (ITS) refer to the integration of advanced technologies and communication systems

into the transportation infrastructure and vehicles. One of the key elements of ITS is the deployment of Roadside Units (RSUs). A Roadside Unit (RSU) is a device used in Vehicular ad hoc networks (VANETs), which is installed on the side of the road or other infrastructure. Its main goal is to provide the vehicles in its surroundings with reliable real-time information about the road conditions and traffic, based on inputs of vehicles, sensors or other sources. This allows drivers to make informed decisions while driving and has the potential to greatly improve the safety and efficiency of the traffic [1].

Security and latency play crucial roles in the whole process of road condition monitoring [2]. First of all, it is of utmost importance that the received messages of the RSUs are reliable and thus are based on inputs of authorized and authenticated vehicles. Also, since vehicles are closely related to people's life, from a privacy point of view, it is critical that an outsider, even a RSU, is not able to track a particular vehicle. This follows from the fact that a movement pattern of a particular vehicle can leak sensitive information about health, religion, sexual orientation, etc. Second, the aspect of real-time is important to properly react on emergency situations. In order to decrease the latency in the process, RSUs should have the capability to handle autonomously, instead of relying on input of a central authority or cloud server. It should act as a fog node in the network [3]. Moreover, compute-intensive cryptographic operations like pairing operations, required to enable the security features, should be avoided if possible [4].

A. Motivation of the Paper

It has been noticed in [5], that there is only limited literature on the VANET-related security schemes in which the RSU is able to handle locally the security checks. Moreover, in the list of existing schemes, many of them are using pairing-based operations or do not offer confidentiality. As a consequence, a vehicle-to-fog (V2F) authentication scheme, called VFAS, based on elliptic curve cryptography (ECC) has been proposed to deal with these shortcomings [5]. VFAS has the lowest communication and computation costs compared to related work. However, we show that the scheme is vulnerable to impersonation attacks, does not offer unlinkability or untraceability, is insecure for perfect forward secrecy and ephemeral key leakage, does not take revocation of certificates into account, has huge scalability issues and is not resistant against a semi-trusted third party.

Manuscript received 20 June 2023; revised 21 October 2023, 23 February 2024, and 22 March 2024; accepted 11 April 2024. Date of publication 17 April 2024; date of current version 19 September 2024. This work was supported in part by the Horizon Europe Marie Skłodowska-Curie Actions under Project TRACE-V2X (Grant Agreement 101131204) and in part by COST Action CA22104 Beingwise + Vlaams Beleidsplan Cybersecurity. 2024-2028. The review of this article was coordinated by Dr. Qing Yang. (Corresponding author: Awaneesh Kumar Yadav.)

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Digital Object Identifier 10.1109/TVT.2024.3390607

In order to propose a viable alternative, we rely on related work of security schemes for client-server-based architectures [6]. In particular, we transform a highly efficient ECC-based scheme of [7] into this context and show that it addresses the required needs and is highly efficient. One of the changes is that we can not take the same registration process, which relies on the Elliptic Curve Qu Vanstone (ECQV) certificate scheme, because it is inherently insecure for perfect forward secrecy and leakage of ephemeral key material [8], making the scheme vulnerable in the Canetti–Krawczyk (CK) and extended Canetti–Krawczyk (eCK) adversary models. These models provide the attacker additional power of having access to either the long-term key material or the temporary session-specific data of each entity in the communication and are applied in the latest generation of security protocols [9], [10], [11].

B. Contributions of the Paper

To summarize, the main contributions of the paper are as follows:

- We show several security and performance issues for a recently proposed authentication and key agreement scheme for V2F communication, called [5]. In particular, the scheme is vulnerable for impersonation attacks, does not offer unlinkability or untraceability, is insecure for perfect forward secrecy and ephemeral key leakage, does not take revocation of certificates into account, has huge scalability issues and is not resistant against a semi-trusted third party.
- We propose an improved version, called iVFAS, which is able to offer a high level of security, for V2F communication.
- The registration phase in our newly proposed scheme also provides a secure version of the ECQV certificate scheme in the eCK adversary model.
- To guarantee the security of the designed protocol, we conduct both informal and formal (using Real-Or-Random (ROR) logic and Scyther validation tool) security verification.
- A comprehensive performance analysis is presented for the newly proposed scheme, comparing it with other related works in the literature. The findings indicate that our proposed scheme excels in terms of communication, computational, storage, and energy consumption costs. Additionally, we evaluate the performance under unknown attacks, demonstrating that our proposed protocol performs better than most of the other related works cited in the literature.
- We implement the designed protocol using NS3, showcasing its real-time applicability.

The paper is structured as follows. Related work is discussed in Section II. Some preliminaries are provided in Section III. We present the issues on VFAS in Section IV and a new scheme iVFAS is proposed in Section V. The security and performance analysis are provided in Section VI and Section VII, respectively. Finally, we conclude the paper in Section VIII.

II. RELATED WORK

When making a classification of the most relevant security-related papers in VANET, we distinguish different criteria. First, the architecture is important, being a fog-based scheme or centralized scheme. In the first type (A1), the fog does the security verification locally, while in the second type (A2) the fog needs to forward (part of) the message to a trusted cloud server who makes the decisions and replies back to the fog with the response, causing a huge delay. Moreover, in particular, in the fog-based architecture (A1), we also see schemes that are addressing either both vehicle-to-infrastructure (V2I) and vehicle-to-vehicle (V2V) or only V2I communication.

The second main category to distinguish is based on the type of cryptographic operations that are used. This varies from only symmetric and hash-based operations (O1), basic public key-based operations like exponentiation and elliptic curve-based operations (O2-O3), to compute-intensive operations like pairing operations (O4). Note that we have made a difference in the public key-based schemes, O2 and O3, following an observation mentioned by [6]. There, it states that the main point of using public key-based operations above symmetric key-based operations is to offer resistance against a semi-trusted entity defining the security material. In symmetric key cryptography, you completely rely on the trust of this entity because it determines the key material and shares it among the different participating entities. This can be avoided thanks to the use of public key cryptography by using constructions as, for instance, in the Qu-Vanstone certificate management [12] or implicit key management systems [13], where the private key is constructed based on secret key material of the participating entity, which is not known by the trusted entity (called O2). If, in a public key-based scheme, the trusted party is just sharing the secret key material with the entities (called O3), the added value of public key cryptography is lost and the more efficient symmetric key cryptography can be used instead [14].

Third, we also look if the schemes are offering confidentiality of the messages or only focus on authentication. In fact, confidentiality should be required if full privacy needs to be guaranteed to its users.

Table I provides an overview of more than thirty relevant papers describing security schemes for the VANET application domain, classified into the categories described above. As can be seen from the table, almost half of the schemes have a fog-based architecture. Approximately, 60% of the schemes, for both types of architectures, are using compute-intensive operations. In most of the schemes that are offering V2V communication, there is no data security included, except for [16], [24], [25]. However, the confidentiality in schemes [24], [25] is established by means of a group key, which is not practical for dynamically changing groups, often the case in traffic. Note that although the scheme of [23] in its original description is not offering confidentiality, the feature can be easily included as a common shared key between the RSU and vehicle has been determined.

As can be concluded from Table I, the list of schemes with a fog-based architecture, relying on simple public key-based operations, and being capable of offering confidentiality in a

TABLE I
COMPARISON OF SECURITY RELATED PAPERS IN VANET WITH RESPECT TO
ARCHITECTURE TYPES A1-A2; COMMUNICATION NETWORKS V2V,V2I;
OPERATIONS O1-O4; CONFIDENTIALITY Y(YES),N(NO)

Ref.	Year	Arch.	Com.	Ops	Conf
[15]	2011	A1	V2I	O4	N
[16]	2012	A1	V2V,V2I	O4	Y
[17]	2013	A1	V2V,V2I	O4	N
[18]	2014	A2	V2I	O4	Y
[19]	2015	A1	V2V,V2I	O2	N
[20]	2016	A1	V2I	O4	Y
[21]	2016	A1	V2V,V2I	O2	N
[22]	2016	A1	V2I	O4	Y
[23]	2017	A1	V2V,V2I	O2	(Y)
[24]	2018	A1	V2V,V2I	O1	Y
[25]	2017	A1	V2V,V2I	O4	Y
[26]	2019	A2	V2I	O4	Y
[27]	2019	A1	V2I	O4	N
[28]	2019	A2	V2I	O1	Y
[29]	2019	A2	V2I	O3	Y
[30]	2020	A1	V2I	O4	Y
[31]	2020	A1	V2V,V2I	O2	N
[32]	2020	A2	V2V,V2I	O2	N
[33]	2020	A2	V2I	O1	N
[34]	2020	A1	V2I	O4	Y
[35]	2021	A2	V2V,V2I	O4	N
[36]	2021	A2	V2I	O4	Y
[37]	2021	A2	V2I	O4	Y
[38]	2021	A2	V2V,V2I	O4	N
[39]	2022	A1	V2V,V2I	O4	N
[40]	2022	A2	V2I	O4	Y
[41]	2022	A2	V2I	O4	N
[42]	2022	A2	V2I	O3	N
[43]	2022	A2	V2I	O4	Y
[44]	2022	A2	V2I	O4	N
[45]	2022	A1	V2I	O1	Y
[46]	2023	A1	V2I	O1	Y
[47]	2023	A1	V2I	O3	Y
[48]	2023	A1	V2I	O3	Y
[5]	2023	A1	V2I	O3	Y

scalable manner, i.e. without group encryption, is highly limited and restricted to [5], [23], [45]. However, from a security point of view, the scheme of [23] is not ideal as it is not offering perfect forward secrecy and does not address revocation. Also, the user-friendliness of the scheme is not straightforward as it includes multifactor authentication, requiring passwords and biometrics to start the process. Moreover, it is also assumed that the public key of the RSU is consulted via a publicly available database, which is not evident to be realized in practice. The scheme of [45] relies on elliptic curve cryptography, but does not offer security against a semi-trusted entity. In addition, it is vulnerable for denial of service attacks, especially in combination with the batch verification. In the next section, we will demonstrate several severe issues with VFAS. Consequently, it is clear that there is a need for alternative schemes addressing V2I communication by means of ECC-based schemes offering confidentiality.

There are also schemes [47], [48] with a fog-based architecture, where the public key-based operations do not offer security against a trusted third party. Both papers [47], [48] are using a similar technique of implicit certificates, but [47] is relying on a Physical Unclonable Function (PUF) and [48] on biometric operations during the authentication process. However, the main issue in both papers is the scalability. The RSU needs to try all public keys of the vehicles to proceed further in the mutual

authentication protocol, which is unpractical for places where many vehicles pass by. In addition, the schemes are not resistant to DoS attacks as the RSU cannot detect the validity of the first request. In addition, [48] does not offer perfect forward secrecy either.

Instead of proposing a new scheme based on small modifications of VFAS [5], which is typically done in literature, we prefer to start from scratch and looked for inspiration in the literature of security schemes, defined with a client-server architecture. Although the rationale of using public key cryptography has been clearly stated in [6] and demonstrated based on a substantial list of papers published in 2019–2020, we still see many recent papers making the same mistake when defining the key material like for instance in [49]. In addition, in almost all of the VANET papers, only the Dolev-Yao (DY) adversary model has been considered, in which the adversary is able to eavesdrop on the communication channel and to delete, alter, change, insert parts of the data sent over the channel. However, nowadays, the more strict Canetti–Krawczyk (CK) and extended Canetti–Krawczyk (eCK) adversary models are being the standard [9], [10], [11]. In [7], an ECC-based authentication and key agreement protocol has been proposed, for which the authentication protocol is resistant in these adversary models, and which offers security features like anonymity and untraceability. However, the registration phase, which was based on the ECQV protocol, is not resistant in the eCK adversary model. Moreover, several other subtle changes are required in order to be applied in a V2I context.

III. PRELIMINARIES

This section covers the concepts that are used in the protocol design.

A. Architecture

In V2I, communication between vehicle and fog is considered, where the fog provides the first analysis and makes the first decisions, before sending the aggregated data to a cloud server [5]. A trusted third party (TTP) or certificate authority (CA) is responsible for the generation of the key material and corresponding certificates. The main goal of the protocol is to establish a secure session key between the vehicle and fog, based on the key material defined with the aid of the TTP. A graphical presentation of this architecture is provided in Fig. 1.

B. Adversary Model

We here consider the DY [50], CK [51], and eCK [52] adversary models. This includes that protection needs to be offered against any passive and active attacker, which is able to eavesdrop on the communication channel and to actively manipulate the message. In addition, it is also assumed that the attacker manages to get access to either the long-term secret key material or the temporary session key data of both entities. With respect to cryptographic assumptions, the security strength of the protocol relies on the hardness of both the elliptic curve discrete logarithm problem (ECDLP) and elliptic curve diffie Hellman problem (ECDHP), as will be later demonstrated in the security analysis of the protocol (see Section VI).

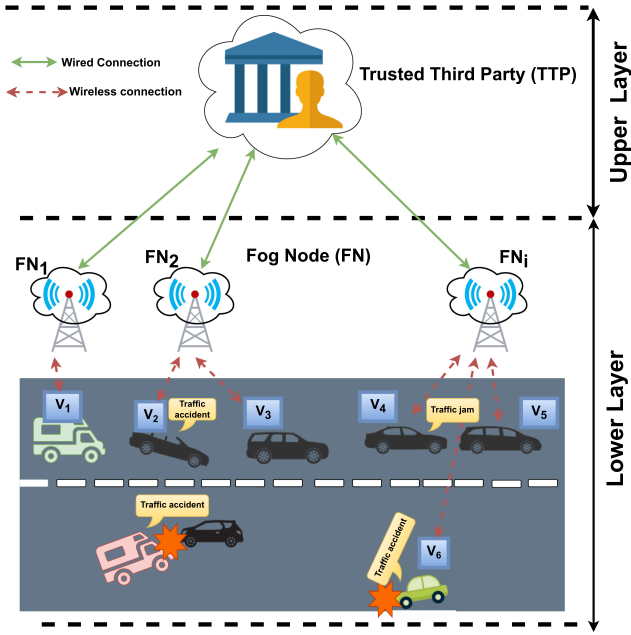


Fig. 1. The system model of the fog-based RCMS in IoV [5].

C. Security Features

In contrast to the limited list of security features mentioned for VFAS [5], we consider the more extended list from [7], which includes all security requirements valid for any IoT-related use case, and thus is also applicable in a V2I domain.

- Mutual authentication, confidentiality, and anonymity are classical features, mentioned also in [5]. In addition, resistance against classical attacks like impersonation, man-in-the-middle, and replay attacks have been included in [5].
- Unlinkability or untraceability is also essential because it avoids linking messages with each other. There have been several approaches mentioned in literature [53] in which it has been possible to retrieve the identity from linking the same message and thus results in breaking the anonymity.
- Leakage of session-specific information is important to take into account as strong random number generators are typically very hard to construct. If in some cases the random values become deterministic, then the previously generated session keys should still remain secure.
- Perfect forward secrecy is essential as it would allow an attacker to reveal all previously generated session keys, and thus to break the confidentiality of previously sent messages, in case the secret key of one of the communication parties is leaked.
- Synchronization attacks are relevant for security protocols in which the identity changes in order to offer protection against unlinkability.
- Complete protection against denial of service attacks is very hard. As a minimum, it is required that the recipient can verify the authenticity of each message in order to avoid storage of material in the buffer until a response is received.
- Resistance against a semi-trusted TTP implies that although the TTP executes the required actions, it might be interested to derive the session key itself or to impersonate the entities in the scheme in order to retrieve the data and to

TABLE II
NOTATIONS AND MEANINGS USED IN VFAS [5]

Notation	Meaning
$\oplus, , h()$	Exclusive or, concatenation, hash function
$(a_i, c_i), (A_i, B_i)$	Private and public key of vehicle
x_i, y_j	Random numbers
RID_i	Real identity of vehicle
P	Generator of elliptic curve (system parameter)
s & P_{pub}	Private and public key of TTP
ID_{fj}	Identity of fog
PID_j	Pseudo-identity of vehicle
$SK_{ij} = SK_{ji}$	Session Key between the V_i and FN_j

sell them for its own purposes. Due to the fact that a lot of personal information is involved in V2I communication, it is important to offer protection against this scenario.

IV. ATTACK ON VFAS

In order to describe the flaws of the scheme, we need to explain the registration, mutual authentication and key agreement phase, and road condition monitoring phases of VFAS as described in [5]. Table II gives an overview of the main notations used in there.

1) *Registration by Vehicle*: First, for the registration of the key material of the vehicle, the TTP generates upon receiving the real identity RID_i and the EC point $A_i = a_iP$, with a_i a randomly chosen value and P a generator of the commonly agreed EC, the tuple (c_i, C_i) . Here $C_i = B_i \oplus h(RID_i)A_i$ with b_i a randomly chosen value for which $B_i = b_iP$ and $c_i = b_i + sh(A_i||B_i)$ where s is the private key of the TTP for which $P_{pub} = sP$ and $h()$ a hash function. The vehicle stores as private key the tuple (a_i, c_i) and as public key the tuple (A_i, B_i) .

2) *Registration by Fog Node*: For the registration of the fog node with identity ID_j , the TTP chooses a random value d_j for which $D_j = d_jP$ and $sk_j = d_j + sh(ID_j||D_j)$ is computed and is sent to the fog node to serve as public and private key respectively.

3) *Mutual Authentication and Key Agreement*: The mutual authentication and key agreement phase is illustrated in Fig. 2. A pseudo-identity PID_i is created based on a randomly chosen x_i on which a signature σ_{i1} is created. This signature is then verified by the fog node and if successful, the fog also chooses a random value y_j , which is used in combination with x_i to create a kind of Diffie-Hellman (DH) key y_jX_i on which a signature is defined σ_{j1} . Upon the arrival of the response of the fog, the vehicle can compute the session key SK_{ij} and verify the signature.

4) *Road Condition Monitoring*: Denote the message with road information by m_i and current timestamp by t_i , then each vehicle sends the message $E_{SK}(m_i, t_i, h(m_i||t_i))$, using its session key SK , to the fog node.

Once receiving this message, the fog can decrypt the message in order to extract the road information message and to verify its correctness and freshness. The message is only accepted if at least a threshold of ω messages with equivalent data on the road condition are received.

B. Description of the Attacks on VFAS

1) *Impersonation of Fog by Vehicle*: With the received key material of the TTP, any vehicle can impersonate the fog by

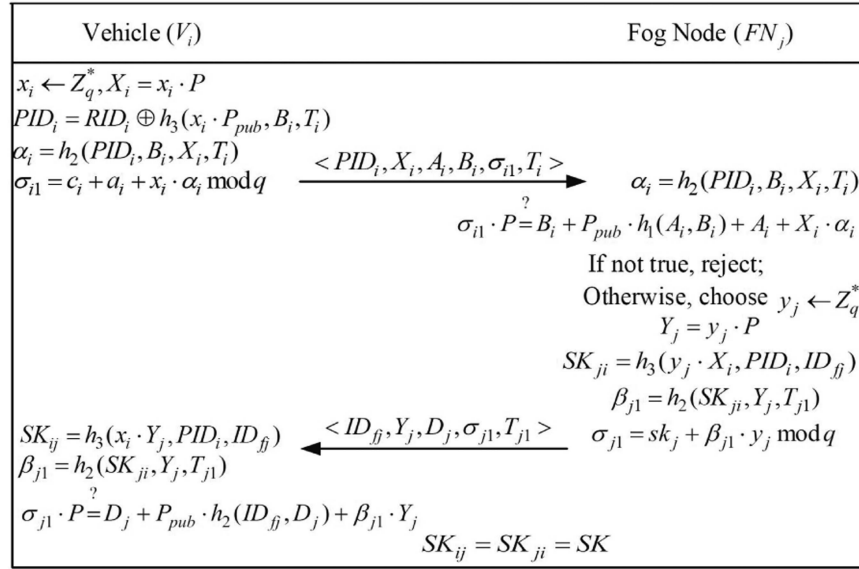


Fig. 2. Authentication and key agreement of VFAS [5].

replacing the identity of the fog ID_{fj} by A_i , the public key D_j by B_i and the private key sk_j by c_i . This follows from the fact that for the vehicle the following relation between the key material is made $c_i = b_i + sh(A_i || B_i)$, which is similar to the construction for the fog $sk_j = d_j + sh(ID_j || D_j)$.

As a consequence, any vehicle can take over the role of the fog and define a session key with the vehicles in the field. There is no way for the vehicle to detect that the received response is coming from a legal fog and thus the authentication of the scheme is broken.

2) *Security Against Unlinkability*: Although the protocol offers protection against identity anonymity, it still allows tracking of the vehicle by anyone sniffing on the channel as long as the same key material is used. This follows from the fact that A_i, B_i are sent in clear text and they remain the same for each message coming from the same vehicle. Moreover, since the vehicle passes the new value of A_i to the fog in any new update request for key material to the TTP, the fog can even further track the vehicle.

In many papers, unlinkability has been considered as an essential feature, because it might allow an attacker to link the real identity of the vehicle with the identity-related information (A_i, B_i) based on pattern recognition.

3) *Perfect Forward Secrecy*: Perfect forward secrecy means that if the long-term key material is revealed, no further information can be retrieved on previously established session keys. Although, while this feature is not claimed to be satisfied in the scheme by the authors, it is an important security feature, which is normally added in any security scheme.

In VFAS, if the private keys (a_i, c_i) of the vehicle are leaked, based on the fact that $\sigma_{i1} = c_i + a_i + x_i \alpha_i$ is sent on the insecure channel and $\alpha_i = h(PID_i || B_i || X_i || T_i)$ is built upon data also sent over the insecure channel, it is possible to derive x_i . Once x_i is leaked, it is easy to also derive $SK_{ij} = H(x_i Y_j || PID_i || ID_{fj})$, as the corresponding missing pieces of the hash are sent over the insecure channel.

4) *Security Against Ephemeral Leakage*: Offering security for leakage of temporarily ephemeral key material is also a feature that was not originally addressed by the authors, but plays an important role in the current generation of security protocols. It implies that if temporarily ephemeral key material is somehow leaked, for instance by using a weak random generator that leaks additional information making it possible to derive the random values in the protocol, then, it should be still impossible to derive the commonly agreed session keys.

By the definition of the session key, it is clear that if the random value at the side of the vehicle, or the random value at the side of the fog is leaked, then the session key can immediately be reconstructed.

5) *Revocation of Certificates*: There is totally no possibility in the scheme to verify if the public key material used, is still valid and not in possession of an attacker. No mention has been made on key revocation mechanisms and so old values of (A_i, B_i) or D_j can be reused at any time.

6) *Scalability*: In the road condition monitoring phase, the vehicle is sending the message $E_{SK}(m_i, T_i, h(m_i, T_i))$. However, the fog node does not know which session key SK it needs to use and so it requires first the storage of the session keys of all the vehicles in order to exhaustively try among the whole list. This is very compute-intensive and has not been taken into account in the performance analysis of the scheme. Of course, this can be easily overcome by including the pseudo-identity PID_i at each message, but again at the cost of losing the untraceability feature (which is in fact already not present).

7) *Protection Against a Semi-Trusted TTP*: The TTP does not possess the full private key material of the vehicle. However, it does possess the private keys of the fogs as they are generated by the TTP and send to them. As a consequence, the TTP can also impersonate the fog. Moreover, since the scheme does not satisfy perfect forward secrecy, it can also derive all session keys and follow the whole communication between vehicles and fog nodes.

TABLE III
NOTATIONS AND MEANINGS USED IN IVFAS

Notation	Meaning
G	Generator of the curve
T_s	Validity period of certificate
(d_i^0, Q_i^0)	Long term private-public key of vehicle
(d_i, Q_i)	Private-public key used in authentication phase
K_1	DH Key
(d_f, Q_f)	Private-public key pair of fog
$a_i^0, b_i^0, r_i, r, r_f$	Random numbers
$H()$	Extendable output hash function

V. PROPOSED SCHEME IVFAS

As a solution to overcome the shortcomings in the VFAS scheme, we propose the iVFAS protocol or improved VFAS protocol consisting of the same phases as VFAS, i.e. initialization phase, registration phase, authentication and key agreement phase, road condition monitoring phase and key update phase.

In particular, the road condition monitoring phase is exactly the same, which just exists of using the commonly agreed session key between vehicles and fog to communicate the road state and if sufficient vehicles are sending the same situation, the information will be validated. However, in order to avoid the scalability issue like in VFAS, we here include a temporary identity in the message, which is valid during one time period defined by the fog. The smaller the time period, the higher the protection against untraceability becomes, taking into account the additional computation and communication costs of each new key agreement procedure.

For the core of the protocol, the authentication and key agreement phase, we will rely on the authentication protocol of [7] presented for an RFID-based use case. In particular, the scheme without storage from the reader side will be considered. The advantage of this scheme is that it is not only highly efficient, it also satisfies the most important security features, which are relevant in this context too.

However, we can not take the same registration process as in [7], which relies on the ECQV certificate scheme and is thus inherently insecure for perfect forward secrecy and leakage of ephemeral key material [8] and requires a secure channel if the anonymity of the user should be guaranteed. In the security analysis of the scheme [7], the security of the registration process was neglected as it was a one-time operation. However, as the vehicle needs to undergo frequent registrations, attention needs to be made for it. In addition, due to this high frequency, we also need to assume an insecure channel, which is not the case in [7]. Since the proposed registration phase is defined over an insecure channel, the key update phase can be similar and so no separate protocol is required for that operation (as in [5]). The main notations (different from VFAS) used in iVFAS are summarized in Table III.

A. Initialization Phase

The system parameters, being elliptic curve (EC), generator of the curve G , the hash function H , symmetric encryption E_K and decryption D_K function with secret key K , public key of TTP P_{Pub} are stored at each entity. Also, T_s is a system parameter, which refers to the validity period of the certificates of the

vehicles in the system and is supposed to be fixed and automatically programmed in each of the vehicles. T_s can represent for instance a specific day, even hour, depending on the required frequency of change. The determination of the length of the period is a trade-off between privacy and performance and is considered outside the scope of this paper. Each vehicle possesses a real identity RID and corresponding public key pair (d_i^0, Q_i^0) , for which the combination (RID, Q_i^0) is publicly available data. Fig. 3 shows the proposed registration phase of vehicle.

B. Registration Phase

We will first explain the mechanism to be applied for the vehicle and then discuss some subtle changes that are applicable to the fog. The process needs to be repeated at least once every T_s . It can also be possible to derive multiple certificates in one T_s , in order to obtain a higher privacy with respect to the same fog. For ease in notation, we only consider one certificate per T_s .

1) *Vehicle*: We here assume that the vehicle possesses a long-term private key d_i^0 , which is stored in the tamper-proof memory of the vehicle. For the generation of a random point $A_i = a_i G$, the scalar a_i is constructed of the sum of the long-term key and a randomly chosen value a_i^0 , i.e. $a_i = d_i^0 + a_i^0$. Next, the DH key $K_1 = a_i P_{pub}$ is derived in order to define the cipher text $C_1 = E_{K_1}(RID_i, A_i, h_i)$, with $h_i = H(d_i^0 P_{pub} || T_s)$. The message $\{C_1, A_i\}$ is sent to the TTP. The TTP starts decrypting C_1 by computing $K_1 = s A_i$, with s the private key of the TTP. As a consequence, the TTP retrieves the real identity and checks if the corresponding hash value is computed by the identity which is in possession of the private key associated with the public key Q_i^0 .

Next, it also computes $B_i = b_i G$, where $b_i = s + b_i^0$, with b_i^0 a randomly chosen value. The key $K_2 = b_i A_i$ is defined. Next, the certificate is now defined as $Cert_i = H(K_1 || K_2)(A_i + B_i)$ and the auxiliary data for the vehicle to compute its key material is defined as $z_i = H(Cert_i | T_s) b_i + s$, which is encrypted as $C_2 = E_{K_2}(z_i)$. The message $\{C_2, B_i\}$ is sent to the TTP. The TTP also stores $(RID_i, Cert_i)$ during the time period of T_s for revocation purposes.

Now, the vehicle can also derive $K_2 = a_i B_i$ in order to retrieve z_i . Next, the vehicle computes the certificate $Cert_i = H(K_1 || K_2)(A_i + B_i)$ and its private key as $d_i = z_i + a_i H(Cert_i || T_s)$. In order to check the validity, it should ensure that $d_i G = H(Cert_i | T_s) Cert_i + P_{pub}$.

2) *Fog*: For the fog, the registration process is similar, except that an identity ID_{fj} is included in the process. Here, the identity refers to the geographic coordinates of the fog, which is publicly available information that can be verified by the vehicles. So, the public key of the fog is now defined by $Q_f = d_f G = H(Cert_f || ID_{fj} || T_s) Cert_f + P_{pub}$.

C. Authentication and Key Agreement Phase

Although the flow and inherent operations are very similar as in [7], we integrate some subtle differences in order to fit it into the use case of V2F communication. These changes are mainly related to the fact that the identity of the vehicle is not included and that we need to define a session key, as the scheme of [7]

Vehicle	TTP
$(d_i^0, Q_i^0), P_{pub}$	(s, P_{pub})
Choose a_i^0 $a_i = d_i^0 + a_i^0$ $A_i = a_i G$ $K_1 = a_i P_{pub}$ $h_i = H(d_i^0 P_{pub} T_s)$ $C_1 = E_{K_1}(RID_i, A_i, h_i)$	$\xrightarrow{C_1, A_i}$ $K_1 = s A_i$ $RID_i, A_i, h_i = D_{K_1}(C_1)$ Verify $h_i = H(s Q_i^0 T_s)$ Choose b_i^0 $b_i = s + b_i^0$ $K_2 = b_i A_i$ $Cert_i = H(K_1 K_2)(A_i + B_i)$ $z_i = H(Cert_i T_s) b_i + s$ $C_2 = E_{K_2}(z_i)$ $\xleftarrow{C_2, B_i}$ Store $(RID_i, Cert_i)$
$K_2 = a_i B_i$ $z_i = D_{K_2}(C_2)$ $Cert_i = H(K_1 K_2)(A_i + B_i)$ $d_i = z_i + a_i H(Cert_i T_s)$ Check $d_i G = H(Cert_i T_s) Cert_i + P_{pub}$	

Fig. 3. Proposed registration phase of vehicle.

Fog	Vehicle
$ID_{fj}, (d_f, Q_f, Cert_f), (cert_{rn})_n, P_{pub}$	$(d_i, Q_i, Cert_i), P_{pub}$
Choose $r_f, R_f = r_f G$ $(Cert_i r) = C_i \oplus (d_f + r_f) R_i$ Check: $Cert_i \in (cert_{rn})_n$ $Q_i = H(Cert_i T_s) Cert_i + P_{pub}$ $H(Q_i R_f R_i r) = (h_0, h_1, h_2)$ Check: $s_i G = R_i - h_2 Q_i$ $SK = h_0$	$\xrightarrow{R_f, ID_{fj}, Cert_f}$ $Q_f = H(Cert_f ID_{fj} T_s) Cert_f + P_{pub}$ Choose r_i, r $R_i = (r_i + d_i) G$ $C_i = (Cert_i r) \oplus (r_i + d_i)(Q_f + R_f)$ $H(Q_i R_f R_i r) = (h_0, h_1, h_2)$ $s_i = (r_i + d_i) - d_i h_2$ $\xleftarrow{R_i, C_i, s_i}$ $\xrightarrow{h_1}$ Check $h_1, SK = h_0$

Fig. 4. Proposed authentication and key agreement between vehicle and fog.

is only an authentication scheme. Fig. 4 shows the proposed authentication and key agreement between vehicle and fog.

To start, we consider that the fog at timestamp T_s is in possession of a valid private-public key pair (d_f, Q_f) with certificate $Cert_f$ and is able to access a list of revoked certificates $cert_{rn}$ of vehicles for the considered period T_s . This list can be locally stored or consulted via an online trusted and integrity protected source like for instance being stored on a distributed ledger [54].

At the vehicle side, the certificate $Cert_i$, together with its key pair (d_i, Q_i) for the given period T_s is available. Unlike in [7]

where the reader is fixed, the fog needs to broadcast, besides the random EC point $R_f = r_f G$, also its identity ID_{fj} and certificate $Cert_f$.

Based on this input, the vehicle is able to retrieve the corresponding public key $Q_f = H(Cert_f || ID_{fj} || T_s) Cert_f + P_{pub}$. Note that the vehicle verifies the validity of the identity with the geographic location. Then, the vehicle generates two random values r_i, r . It then defines a random EC point $R_i = (r_i + d_i) G$. Next, a DH key is made with data combining both the ephemeral and private key material in order

to hide its certificate and a random value r into the variable C_i , i.e. $C_i = (Cert_i || r) \oplus (r_i + d_i)(Q_f + R_f)$. Next, it defines $H(Q_i || R_f || R_i || r) = (h_0, h_1, h_2)$, where h_0, h_1, h_2 are the outcomes of an extendable hash function like e.g. SHAKE128. These components are used as session key and to verify the authentication.

Note that no session key was determined in [7]. As such, h_2 is included in the signature $s_i = (r_i + d_i) - d_i h_2$. The message $\{R_i, C_i, s_i\}$ is then sent to the fog. Note that the addition of the parameter r is required in order to guarantee anonymity in case $Cert_i$ is leaked (e.g. put on the revoked list of certificates) and thus the signature corresponding with $Cert_i$ could be verified.

At the fog side, by computing the DH key $(d_f + r_f)R_i$, it can retrieve the certificate $Cert_i$ and random value r from C_i . The fog then checks if the certificate is revoked and if not, it computes the corresponding public key Q_i by $H(Cert_i || T_s)Cert_i + P_{pub}$. It then verifies the validity of the signature s_i by first also computing the hash $H(Q_i || R_f || R_i || r) = (h_0, h_1, h_2)$. If $s_i G = R_i - h_2 Q_i$, then the request is considered valid, the session key becomes h_0 and the response to prove its authenticity will be h_1 .

Upon arrival of the value h_1 , the vehicle knows that the fog has the required credentials to guarantee its authenticity and that it was able to retrieve the session key h_0 .

In order to link the session key with future messages coming from the vehicle, the fog needs to store the session key SK corresponding with $(H(r || i), (H(r || i + 1), r, i))$. The vehicle communicates with the fog using the identity related data $(H(r), i)$, which is after successful reply of the TTP updated to $(H(r || i + 1), H(r || i + 2), r, i + 1)$. Similar, at the fog side, after receiving a message with $H(r || i)$, the memory is updated to $(H(r || i + 1), H(r || i + 2), r, i + 1)$. Note that both values are required in order to avoid desynchronization issues.

D. Key Refreshment Phase

If T_s is expired, a new key refreshment is required. As suggested in VFAS, key refreshment can be sent directly to the TTP or via the fog. Here, the registration process can be utilized for this purpose.

VI. SECURITY ANALYSIS

We will focus the security analysis on the registration phase, since the security analysis of the authentication and key agreement phase is similar to a large extent to the one of [7] for which both formally and informally the security features mentioned in Section III have been proven. We will consider both an informal and formal security analysis.

A. Informal Security Analysis

We will show that the registration process satisfies the same security features as the authentication and key agreement scheme, corresponding to the required list of security features as provided in Section III.

- *Mutual authentication*: The TTP is able to check the authentication of the vehicle as it can verify the correctness of h_i . Only the entity who is in possession of the private key d_i^0 (corresponding with the public key associated to

RID_i) is able to derive a valid value. Also the vehicle is able to verify the authenticity of the TTP by checking the validity of the derived private key d_i for which $d_i G$ should be equal to $H(Cert_i || T_s)Cert_i + P_{pub}$. As a consequence, the scheme is resistant against man-in-the-middle attacks and impersonation attacks.

- *Anonymity*: An attacker intercepting the messages on the channel will not retrieve information on the identity of the sender/receiver as the messages A_i, B_i are random points on the curve and the messages C_1, C_2 are encrypted using the private key of each of the entities.
- *Unlinkability*: Due to the fact that A_i, B_i (which have also an impact on the definition of C_1, C_2) are random values and are changed in each call, there is no relation between different registration requests.
- *Session specific information*: At the vehicle side, session specific information is a_i^0 , while at the TTP side, this is defined by b_i^0 . None of the DH keys utilized in the scheme can be derived with knowledge of solely these values.
- *Perfect forward secrecy*: If the long term key material s of the TTP is leaked, the anonymity of the vehicle in the request is lost. However, it will not lead to the leakage of the key K_2 , which is needed to make the link with the certificate of the vehicle afterwards and to decrypt C_2 . If the long term key material of the vehicle (d_i^0, d_i) is leaked, the keys K_1, K_2 cannot be revealed. As such, z_i remains secret and thus no further information can be obtained.
- *Replay attacks*: Due to the fact that random values A_i, B_i , which are considered to be used as nonces, are included in the protocol, there is no replay possible. Even if the vehicle would replay the same random value, the TTP will choose a different one and so, the attacker will not be able to continue the rest of the protocol in a successful manner.
- *Synchronization attack*: Due to the fact that there is no change in the internal key material (d_i^0, s), synchronization attacks are not applicable.
- *Denial of service attacks*: As the protocol is a simple challenge-response protocol, in which the participant can immediately verify the validity of the message and thus no buffers are required to store the temporary requests, the efforts for a denial of service attack are not evident.
- *Resistant to a curious TTP*: The TTP is not able to derive the private key itself due to the fact that ephemeral secret key material, generated by the vehicle is required in the definition of the private key.
- *eCK security*: We have already shown that the scheme provides perfect forward secrecy and ephemeral leakage protection. However, in the eCK model, it is also assumed that the scheme should be resistant in case the attacker possesses a combination of either long-term key material or session-specific material in each entity. As such, the attacker possesses in situation (1): (r_f, d_i) or situation (2): $(d_f, (r_i, r))$. In the first case, it is impossible to find SK, as it requires knowledge of r , for which d_f or r_i need to be known and this is impossible due to the hardness of ECDLP. This is similar for the second case, where knowledge of either d_i or r_f is required.

B. Formal Security Analysis Using Scyther Validation Tool

The Scyther validation tool is adopted to ensure that our proposed authentication and key agreement between the vehicle and fog is secure and there is no attack possible. It is a widely accepted validation tool used by many security researchers, especially for authentication protocols. Under this tool, the protocols are specified in the specification description language (.spdl). The security properties of the authentication are validated using the five specified factors by the scyther tool. The description of the five factors is as follows [55].

- *Alive*: guarantees that the parties involved in the protocol perform all communication events.
- *Weakagree*: ensures protection against impersonation and MiTM attacks
- *Nisynch*: guarantees that all messages are sent by one party and received by the other party, and thus offers protection against replay and DoS attack.
- *Niagree*: guarantees that modification will not affect the next phase of the authentication protocol, and thus offers protection against desynchronization attacks.
- *Secret*: ensures that the secret parameters are not known by the adversary.

During the simulation of the registration phase and the authentication phase, Dolev, CK, and eCK threat models are considered. The tool supports all these types of threat models, while most of the other tools (Avispa, ProVerif, etc.) only use the Dolev model. In addition, it provides a graphical view if the attack happens on the protocol. These features make the scyther tool more efficient as compared to others [56].

To verify the specified security features, there are four types of parameters given by the tool. With the first type of parameter (i.e., verification parameters), we check for all matching types such as typed matching and basic types flaws in order to find all type flaws by running them 100 times (maximum). For the second type of parameter (i.e., Adversary model), we take assumptions on the type of attacker (see Dolev, CK, and eCK threat model), also ensuring perfect forward secrecy, reveal session key, random numbers and state (i.e., to assure the ephemeral secret leakage). With the third parameter (i.e., Advanced parameter), we check for all three matching types such as find all attacks, find first attack, and find best attack by running them 100 times (maximum). The last parameter defines the graph output parameter, which is set to the maximum of 32.

We checked the protocol and found that both the registration and authentication protocol pass all the specified security features, indicating that they are free from the attacks as mentioned in Section III-C. An example of the output of the tool for a particular choice of parameters is provided in Fig. 5.

C. Formal Verification Using the Real-or-Random (ROR) Logic

The formal verification of the proposed iVFAS is done using the ROR logic [57]. RoR logic is vastly used by many security researchers to examine the security of authentication protocols. Using this tool, we show that the attacker (A) can not determine the session key of the iVFAS. The proof of the logic involves

The screenshot shows the Scyther tool interface. On the left, the 'Verification parameters' section is visible, including 'Maximum number of runs (0 disables bound)' set to 100, 'Matching type' set to 'Find all type flaws', and 'Adversary compromise model' with options for 'Long-term Key Reveal' (None, Others (DIT), Actor (NCC)) and 'Session-Key Reveal' (Random, State, Automatically infer local state). The 'Advanced parameters' section includes 'Search pruning' (Find best attack) and 'Maximum number of patterns per claim' (100). The 'Graph output parameters' section shows 'Attack graph font size (in points)' set to 32.

On the right, the 'Scyther results: verify' window displays a table with the following data:

Claim	Status	Comments
Authentication_Fog	Ok	No attacks within bounds.
Authentication_Fog1	Nisynch	Ok
Authentication_Fog2	Niagree	Ok
Authentication_Fog3	Secret_h_0	Ok
Authentication_Fog4	Weakagree	Ok
Authentication_Fog5	Alive	Ok
Authentication_Fog6	Secret_h_2	Ok
Vehicle	Authentication_Vehicle1	Nisynch
Authentication_Vehicle2	Niagree	Ok
Authentication_Vehicle3	Secret_h_2	Ok
Authentication_Vehicle4	Weakagree	Ok
Authentication_Vehicle5	Alive	Ok
Authentication_Vehicle6	Secret_h_0	Ok

Fig. 5. Output of the Scyther tool for the registration phase of iVFAS.

two participants, namely probabilistic-polynomial time Turing machines, known as challenger C_H and attacker A . C_H is considered as the owner of the system where the iVFAS executes a game played by C_H and A in which C_H verifies that A can not get any secret through which he/she may determine the secrets of the authentication session. A uses predefined queries to extract the secrets of the authentication and key agreement session. A random oracle model is used in this process. The proposed iVFAS involves *Vehicle*, *Fog*, and *TTP* during the authentication session. In the ROR model, instances of *Vehicle*, *Fog* and *TTP* are denoted as $Vehicle^i$, Fog^j and TTP^k . The proposed iVFAS is run between *Vehicle* and *Fog* using the insecure public channel. A runs the following queries to gain secrets in iVFAS.

- $Execute(Vehicle^i, Fog^j)$: It is run to gain the exchanged message between *Vehicle* and *Fog* during the authentication and key agreement session.
- $Reveal(E^l)$: It is run to gain the session key generated between *Vehicle* and *Fog* during the authentication and key agreement session.
- $Send(E^l, m)$: It is run to gain the response of the replied message from the *Vehicle* and *Fog*.
- $Test(E^l)$: It is run to gain the session key verification by flipping a coin. A coin is tossed, and based on the outcome, the decision is adopted.

Theorem 1: If A tries to gain the session key derived between *Vehicle* and *Fog* within polynomial time, the advantage of A (Adv_A) is bounded by $Adv_A \leq \frac{A^2}{2B} + \frac{(C+D)^2}{F} + 2A_A^{ECDDH}$, where D , C , A , B , and F denote the count of execute queries, send queries, hash queries, range space of a random number, and length of the output from the hash function respectively.

Proof: Under this theorem, A will play the game named $Game_i$ where i can take on values 0, 1, or 2 by running the queries defined above. Let us assume that $Success_{AGame_i}$ represents the event in which the prediction made by A about the random bit c in the $Game_i$ is correct and $Pr[Success_{AGame_i}]$ denotes the probability that A wins $Game_i$. A executes the three games in the following way:

Game ($Game_0$): Under this game, A launches the real time attack. Before the game starts, the selection of the c bit is made by

A. Therefore, the advantage of the adversary from the semantic security is computed as

$$Adv_A = |2Pr[Success_{AGame_0}] - 1| \quad (1)$$

Game (Game₁): Under this game, the aim of *A* is to gain the exchanged messages between *Vehicle* and *Fog* by running the *Execute* query. Afterwards, *A* runs the *Test* and *Reveal* queries to gain the long-term secrets ($Cert_f, d_f, ID_{fi}, d_i, Cert_i$) used in between *Vehicle* and *Fog* during the authentication to determine the session key. However, these secrets are not shared in plaintext between *Vehicle* and *Fog*, but by using the hash function. So, the winning chance of *Game₁* will be identical to *Game₀*.

$$Pr[Success_{AGame_1}] = Pr[Success_{AGame_0}] \quad (2)$$

Game (Game₂): Since the execution of the previous game gives nothing, it can help *A* to determine the session key. Now *A* executes a *Send* query to extract the random numbers (r_f, r_i, r). These random numbers are used in the session key generation and are protected by the hardness of the discrete logarithm problem and the collision resistance property of the hash function. Consequently, *A* can not gain these random numbers even if R_f, R_i is captured. Except for adding random numbers and hash collisions, *Game₂* will be identical to *Game₁*. Consequently, by using the birthday paradox, we can arrive at the following conclusion.

$$\begin{aligned} &Pr[Success_{AGame_1}] - Pr[Success_{AGame_2}] \\ &\leq \frac{A^2}{2^{B+1}} + \frac{(C+D)^2}{2F} + A_A^{ECDDH} \end{aligned} \quad (3)$$

After playing all three games, it is expected that *A* will guess the correct bit

$$Pr[Success_{AGame_2}] = \frac{1}{2} \quad (4)$$

By using (1), 2 and 4, we get

$$\begin{aligned} Adv_A &= |2Pr[Success_{AGame_0}] - 1| \\ \frac{1}{2}Adv_A &= |Pr[Success_{AGame_0}] - \frac{1}{2}| \\ &= Pr[Success_{AGame_1}] - Pr[Success_{AGame_2}] \end{aligned} \quad (5)$$

By using (3) and 5, we get

$$\begin{aligned} \frac{1}{2}Adv_A &\leq \frac{A^2}{2^{B+1}} + \frac{(C+D)^2}{2F} + A_A^{ECDDH} \\ Adv_{AD} &\leq \frac{A^2}{2^B} + \frac{(C+D)^2}{F} + 2A_A^{ECDDH} \end{aligned} \quad (6)$$

The obtained outcome indicates that *A* failed to get any secrets that can give a clue to derive the session key.

VII. PERFORMANCE ANALYSIS

For the performance analysis, we will compare our work with VFAS [5], together with the other schemes also used in the comparison of the VFAS paper [22], [29], [30], [34], and two recently proposed schemes [45], [46]. We did not include the two latest schemes [47], [48] in the comparison analysis as they do not offer scalability and give thus no correct values for the

TABLE IV
COMPARISON OF SECURITY FEATURES

	F_1	F_2	F_3	F_4	F_5	F_6	F_7	F_8	F_9
[22]	×	×	×	✓	×	✓	✓	×	×
[30]	✓	×	×	✓	×	×	✓	×	×
[34]	✓	×	×	✓	×	×	✓	×	×
[45]	✓	✓	✓	✓	✓	×	✓	×	×
[46]	✓	✓	✓	✓	×	×	✓	×	×
VFAS [5]	✓	×	×	✓	×	×	✓	×	×
iVFAS	✓	✓	✓	✓	✓	✓	✓	✓	✓

Notes: ✓ : Secure; ×: Insecure; F_1 : Mutual Authentication; F_2 : Anonymity; F_3 : Unlinkability; F_4 : Session specific information; F_5 : Perfect forward secrecy; F_6 : Replay Attack Protection; F_7 : Synchronization Attack Protection; F_8 : DoS Attack Protection; F_9 : Resistant to curious TTP;

performance as in reality a whole list of potential public keys needs to be tested to successfully finalize the protocol. We will discuss communication, computation, memory aspects and the serving capability of the fog.

A. Security Features Analysis and Comparison

In Section VI, we showed that iVFAS offers all the security features mentioned in Section III. The security analysis done in [58], [59] shows that [22] does not offer mutual authentication, anonymity, and is prone to unlinkability and DoS attacks. The security analysis carried out in [60], [61], [62] shows that [29] is prone to ephemeral secret leakage (ESL) and also does not offer unlinkability and anonymity. The security verification in [63], [64] indicates that [30] is vulnerable to replay attack, DoS, and does not offer unlinkability. The security verification in [34], [65] depicts that [34] is prone to DoS, ESL and does not fully offer mutual authentication. Both [45], [46] do not satisfy protection against a curious TTP and are not resilient to DoS attacks as the fog is not able to verify the correctness of the message after the first pass. For the VFAS [5], we have informally proven in Section IV-B that it is vulnerable to impersonation, session unlinkability, ESL, and also does not offer protection for a malicious TTP and perfect forward secrecy. Therefore, from the security comparison shown in Table IV, we can conclude that our proposed iVFAS offers all the features, while [5], [29], [30], [34], [45], [46] lack multiple ones.

B. Communication Costs

The communication cost between our scheme and the other related work [22], [29], [30], [34], [45], [46], [5] for the authentication and key agreement phase, considering a 128-bit security level, is presented in Table V. Our scheme iVFAS clearly outperforms the other schemes with at least 32%, which corresponds with the second best scheme VFAS.

To be more specific, in our scheme, the response message of vehicle to fog equals 896 bits (i.e., $\langle R_f, ID_{fj}, Cert_f \rangle$), while from fog to vehicle one message of 640 bits (i.e., $\langle R_i, C_i, s_i \rangle$) is constant and a dedicated response of 128 bits (i.e., $\langle h_1 \rangle$) per vehicle is required.

Fig. 6 shows the bandwidth differences between our scheme and VFAS for varying vehicles, from 1 to 200. However, it should be mentioned that in iVFAS there are three communication passes instead of two in VFAS, which might lead to a slightly higher communication latency. The exact impact is difficult

TABLE V
COMPARISON OF COMPUTATION, COMMUNICATION, STORAGE COSTS, ENERGY CONSUMPTION FOR AUTHENTICATION PROTOCOLS

Protocol	Computational cost (ms)				Communication cost Total bits	Storage Cost			Energy Consumption (mj) Time (mj)
	Vehicle side	Fog node side	Total operations	Total time (ms)		Vehicle side	Fog node side	Total bits	
[22]	$6T_{bp}^m + T_H$	$3T_{bp}^m + T_{bp} + T_H$	$9T_{bp}^m + T_{bp} + 2T_H$	8.72	8192	3056	2896	5952	340.5
[29]	$7T_{ecc}^m + 8T_H$	$7T_{ecc}^m + 7T_H$	$14T_{ecc}^m + 15T_H$	0.29	5024	2048	2256	4304	130.8
[30]	$3T_{bp}^m + T_{bp} + 4T_H + T_{ASE}$	$6T_{bp}^m + 5T_H + T_{ASE}$	$9T_{bp}^m + T_{bp} + 9T_H + 2T_{ASE}$	7.93	4256	2856	2796	5652	337.81
[34]	$3T_{bp}^m + T_{bp} + 2T_H$	$5T_{bp}^m + T_{bp} + 2T_H$	$8T_{bp}^m + 2T_{bp} + 4T_H$	10.99	4128	3124	2996	6120	352.8
[45]	$7T_{ecc}^m + 2T_{ecc}^a + 4T_H$	$6T_{ecc}^m + 2T_{ecc}^a + 4T_H$	$13T_{ecc}^m + 4T_{ecc}^a + 8T_H$	0.27	2816	1024	512	1536	139.4
[46]	$18T_H$	$13T_H$	$31T_H$	0.0031	4096	1836	1680	3516	2.71
VFAS [5]	$6T_{ecc}^m + 2T_{ecc}^a + 3T_H$	$5T_{ecc}^m + 3T_{ecc}^a + 4T_H$	$11T_{ecc}^m + 5T_{ecc}^a + 7T_H$	0.23	2240	2592	512	3104	125.58
iVFAS	$3T_{ecc}^m + 2T_{ecc}^a + 5T_H$	$5T_{ecc}^m + 2T_{ecc}^a + 4T_H$	$8T_{ecc}^m + 4T_{ecc}^a + 9T_H$	0.16	1536	2560	2720	5320	93.01

Notes: P: Protocols; Cryptographic operations and execution time [5]: T_{ecc}^m : EC multiplication-0.0200 ms, T_{ecc}^a : EC addition- 0.0014 ms, T_{bp} : bilinear pairing computation- 3.5649 ms, T_{bp}^m : bilinear pairing based multiplication-0.4600 ms, T_{bp}^a : bilinear pairing based addition-0.0500 ms, T_h : one-way hash function-0.0001 ms, T_{ASE} : AES encryption/decryption-0.1300 ms, , Size of secrets recommended by NIST (bits) [67]: T_h :256, T_{ecc} : 256, T_{AES} : 128, random number, identity, timestamp-256. The energy required for cryptographic operations (mj) [68]: T_{ecc}^m : 9.1, T_{bp} : 47, T_{bp}^m : 32, T_{bp}^a : 17.3, T_H : 0.000108, T_{ecc}^a : 4.8, T_{AES} : 0.00217.

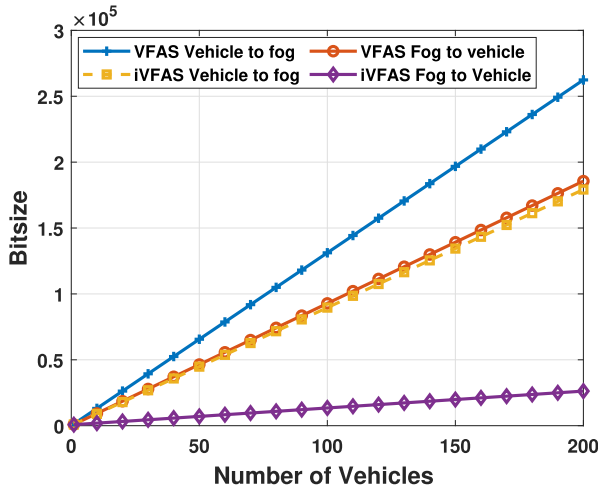


Fig. 6. Bandwidth in our scheme, compared to VFAS.

to assess as it depends of multiple factors, but is typically considered negligible in vehicular IoT communication [67].

C. Computational Costs

The computational costs of the authentication and key agreement phase in our scheme are again drastically lower in our scheme compared to VFAS and most of the other related work, especially at the side of the vehicle (Table V). For instance, for the most compute-intensive operation, being the EC multiplication (EM), we need in total 8 operations, while VFAS requires 11 operations. In particular, at the vehicle side, only 3 multiplications are required, compared to 6 in VFAS. Table V provides a more detailed comparison of the most compute-intensive operations in our scheme. We here distinguish the EC multiplication (EM), the EC addition (EA) and hash operation H. Taking into account the average computational costs for EM, EA and hash operations being 0.02, 0.014 and 0.001 respectively on an Intel Core i5 - 8300 processor, as determined in [5], this results in a cost at the vehicle side which is almost half as big compared to VFAS. Table V represents the computational costs for both our scheme, VFAS and the other related work [22], [29], [30], [34],

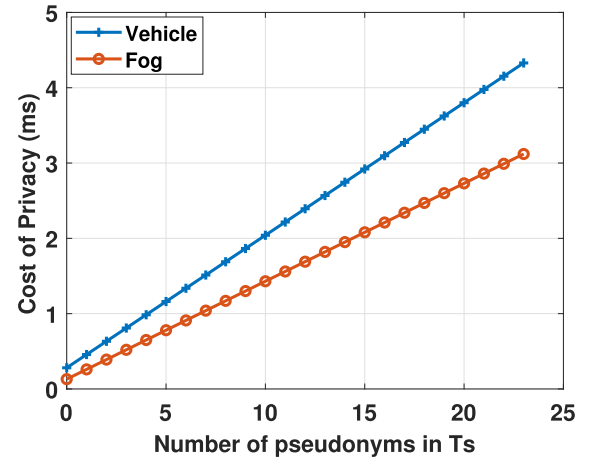


Fig. 7. Cost of privacy for using more pseudonyms in one time period T_s .

[45], [46]. It can be seen that [46] clearly outperforms, which is logic as it is based solely on symmetric key based operations. Still iVFAS, followed by VFAS, outperforms all other schemes.

D. Additional Costs for Privacy Protection

In order to avoid traceability of a vehicle by the fog in one time period T_s , the vehicle can also request multiple m certificates at the same time at the TTP, resulting in a computational cost of $(4 + 3m)EM + (1 + m)EA + (1 + 2m)H + (1 + m)E$ at the vehicle side. In addition, the amount of authentication and key agreement protocols increases with the number of used certificates by the vehicle. For instance, if each vehicle is using the m requested certificates, an additional cost of $(3EM + 2EA + 2H)m$ at the vehicle side and $(5EM + 2EA + 2H)m$ at the fog side is required during a time period T_s .

Fig. 7 shows the evolution of additional computational cost in the function of increased use of certificates or pseudonyms by the vehicle. For instance, if the vehicle wants to change its certificate every hour ($m = 24$) in a time period T_s of one day, a total cost of approximately 4.3 ms at the vehicle side and 3.1 ms at fog side is required.

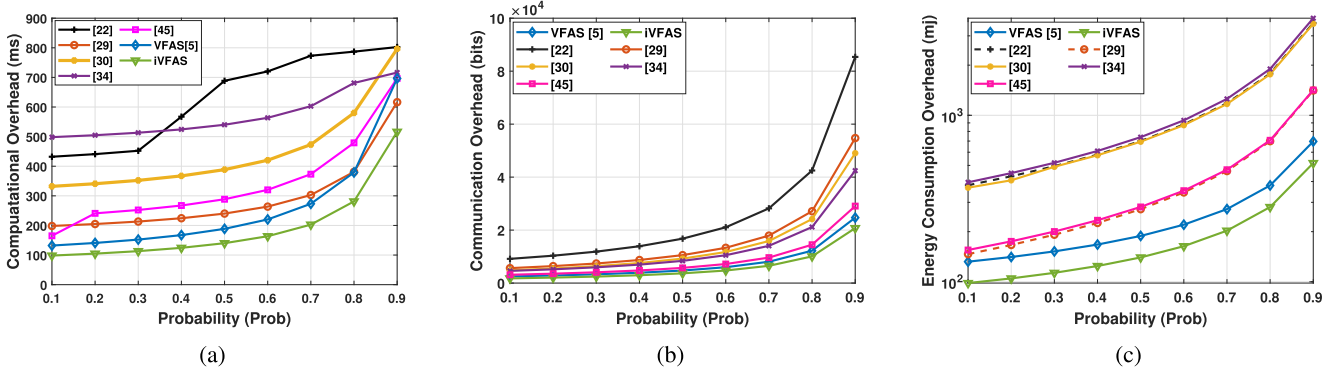


Fig. 8. Comparison of (a) computational, (b) communication, (c) energy consumption overhead under an unknown attack of proposed iVFAS with existing authentication protocols.

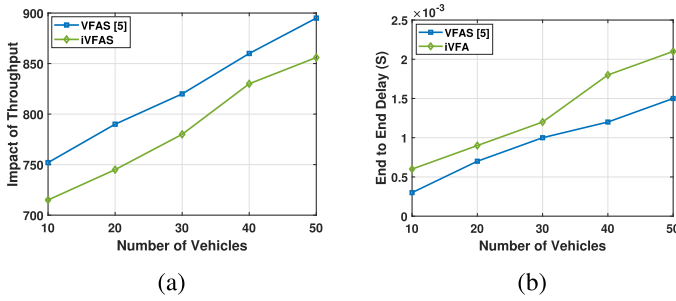


Fig. 9. Comparison of (a) impact of Throughput, (b) end-2-end delay of proposed iVFAS with VFAS.

E. Memory Requirements At Fog Side

At the fog side, the list of revoked certificates needs to be stored in any case and updated every new time period T_s , which corresponds with the validation period of a certificate. This list is communicated by the cloud server via the certificate authority. For the rest, there are two options for storage at the fog side.

In the first option, the fog does not store any key material and for each new communication request, the vehicle needs to go through the authentication and key agreement process. Given the fact that one authentication and key agreement process only requires 4.1 ms, this is a feasible option, even if multiple vehicles send simultaneously a request.

The second option requires the fog to store the tuple $(H(r||i), H(r||i+1), r, i, SK)$ per vehicle, which corresponds to 112 bytes for a session key to be used at most 128 times. If the system limits the storage up to a certain period or up to a usage i of 128 times and forces to restart a new authentication and key agreement process, the amount of required storage is also limited, even for fogs dealing with many vehicles. For instance, taking into account a Raspberry Pi 4 with memory of 8 GB, it would theoretically correspond to a storage of more than 71 million vehicles.

F. Serving Capability At Fog Side

The authentication and key agreement process, together with an authenticated encryption afterwards only requires approximately 0.13 ms at fog side, as determined before. Note that

we here assume that the network communication time and the look-up time in the list of revoked certificates by the fog is negligible.

This implies that during a time interval of 300 ms (considered to be close to the perception of being realtime), the fog is able to handle more than 2200 vehicles. Moreover, if option two is applied where key material is already stored for previously authenticated cars, the number of messages to be dealt with can be even much larger as the authentication and key agreement process can be skipped. Given the fact that the vehicle density typically varies from 600 to 800 on urban roads [5], it clearly demonstrates the feasibility of our proposed scheme.

G. Energy Consumption

This section computes and compares the energy consumed by the fog and the vehicle during the authentication and key agreement process with its competitors. The energy is computed the same way as in [67], which depicts that all the operations are executed on “Strong ARM” CPU running at 133 MHz doing various tasks in order to compute the energy of the operations. In our proposed iVFAS protocol, the energy consumption equals to $1536 \times 0.00066 + 8 \times 9.1 + 4 \times 4.8 + 9 \times 0.000108$, which corresponds to 93.01 mj. The comparison outcome demonstrated in Table V shows that iVFAS consumes much less energy than the others, except for the symmetric key based protocol [46]. The reason to consume less energy is that iVFAS requires less elliptic curve multiplication and addition while VFAS requires more elliptic curve multiplication and addition operations and other uses bilinear pairing.

H. Performance Under Unknown Attacks

In this part, the cost of an unidentified attack on the authentication and key agreement phase is estimated and compared with existing works. We believe that other sorts of attacks might be able to disrupt the performance of the authentication and key agreement process, and it is difficult to predict when they will occur. Despite the fact that the authentication phase is proven to formally and informally prevent all types of assaults, as indicated in Section III-C. Consequently, based on this supposition, we assess the performance and probabilistic impact

TABLE VI
PARAMETERS USED IN NETWORK SIMULATION

Parameters	Value
Operating System	Ubuntu 18.04.6 LTS
Simulator	NS 3.29
Simulator area	1000 * 1000 m ²
Range of FNs	250 m
Number of Vehicles	10, 20, 30, 40 and 50
Number of FNs	25
Mobility	25mps for all scenarios
Transmitting range	300 m
Routing Protocol	Ad Hoc On-Demand Distance Vector (AODV)
Wireless channel bandwidth	6Mbps
Simulation Time	1800s
Wireless protocol	802.11p
Mobility model	Random Direction 2d Mobility Model

on the authentication and key agreement process in terms of computing, communication, and energy usage, where unknown attacks occur.

$$Avg_{cost} = \frac{Success_{cost} \times (1 - Prob) + Failure_{cost} \times Prob}{(1 - Prob)} \quad (7)$$

The above equation (7) is used to estimate the probabilistic impact on performance when an unidentified attack happens in the authentication and key agreement process. The term used in the above equation, such as Avg_{cost} , $Success_{cost}$, $1 - Prob$ and $Failure_{cost}$ represents average cost, successful cost, probabilistic impact and cost of failure when the unidentified attack occurs. Since this attack happens randomly (i.e., the chance of an unknown attack happening in step j is $1/n$, where n signifies the total number of communication messages in a single run of the authentication protocol), and the computation of $Failure_{cost}$ is shown in (8)), before the step j . The outcome shown in Fig. 8(a)–(c) shows the iVFAS not only perform better for known attacks but also for unknown attacks.

$$Failure_{cost} = \sum_{j=1}^n Cost_j * \frac{1}{n} \quad (8)$$

I. Practical Implementation Using NS3

This section presents a comparison of the practical implementations of iVFAS and VFAS using the NS3 network simulator tool. NS3 is widely utilized to demonstrate the practical feasibility of the designed protocols, as noted in previous works [68], [69]. The simulation parameters employed in this study are detailed in Table VI. The simulations are conducted for both iVFAS and VFAS. Following the simulations, a comparative analysis is performed concerning the end-to-end delay and throughput. It is important to highlight that VFAS involves two message exchanges with the transmission of 2240 bits, whereas iVFAS requires three message exchanges and involves the transmission of 1536 bits during authentication.

1) *Impact on Throughput*: This section determines the impact of the output for the VFAS and the iVFAS using Equation (9).

$$E_T = \sum \frac{R_{Pi} \times L_{Pi}}{T}, \quad (9)$$

The symbols used in the equation, namely E_T , R_{Pi} , L_{Pi} , and T , represent the factors related to throughput, packet length for type i , received packet quantity, and total time, respectively. The simulation results for VFAS and iVFAS are presented in Fig. 8. The findings suggest that iVFAS exhibits lower throughput due to the necessity of three message exchanges, whereas VFAS only requires two. However, it is worth noting that VFAS is susceptible to various attacks, and the difference in throughput impact is not substantial. Moreover, the throughput data reveals that as the number of vehicles increases, throughput also rises, primarily due to the increased volume of vehicle and fog authentications.

2) *End-to-End Delay*: This section conducts a comparative analysis to estimate the End-to-End (E2E) delay for both VFAS and iVFAS. In the process of estimating the E2E delay, we calculate the time required for the transmission of packets from the vehicle to the fog node. The End-to-End (E2E) delay for VFAS, iVFAS and its competitors is assessed utilizing Equation 10.

$$E_D = \sum_{k=1}^n \frac{(P_r^k - P_s^k)}{n} \quad (10)$$

The terms employed in Equation 10, namely E_D , P_r^k , and P_s^k , represent the end-to-end delay, time for receiving, and sending of packets of type k , respectively. The findings indicate that iVFAS experiences a higher E_D due to the requirement of three message exchanges, whereas VFAS only necessitates two. However, it is important to note that VFAS is vulnerable to various attacks, and the disparity in E_D is not significant. The results of the end-to-end delay analysis demonstrate that as the number of nodes increases, the end-to-end delay also increases.

From the comparison outcome shown in Fig. 9(a) and (b), we can conclude that VFAS offers better throughput and lower end-to-end delay since it involves two message exchanges, whereas iVFAS requires three message exchanges. However, it is worth noting that VFAS is vulnerable to several attacks, whereas iVFAS is resistant to these attacks. Additionally, iVFAS is lightweight regarding computational, communication, and energy consumption. Therefore, we believe that iVFAS is superior in cost and security compared to VFAS.

VIII. CONCLUSION

This paper presents a very efficient, scalable, and secure security solution for V2I communication. At the same time, the registration phase in the system represents a variant of the ECQV certificate scheme, which is resistant to the eCK adversary model and thus can serve as the basis for many other security protocols. Our scheme can be extended to V2V communication. However, this requires that each vehicle know the list of revoked certificates. In future work, we will investigate the feasibility and impact of applying blockchain technology.

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